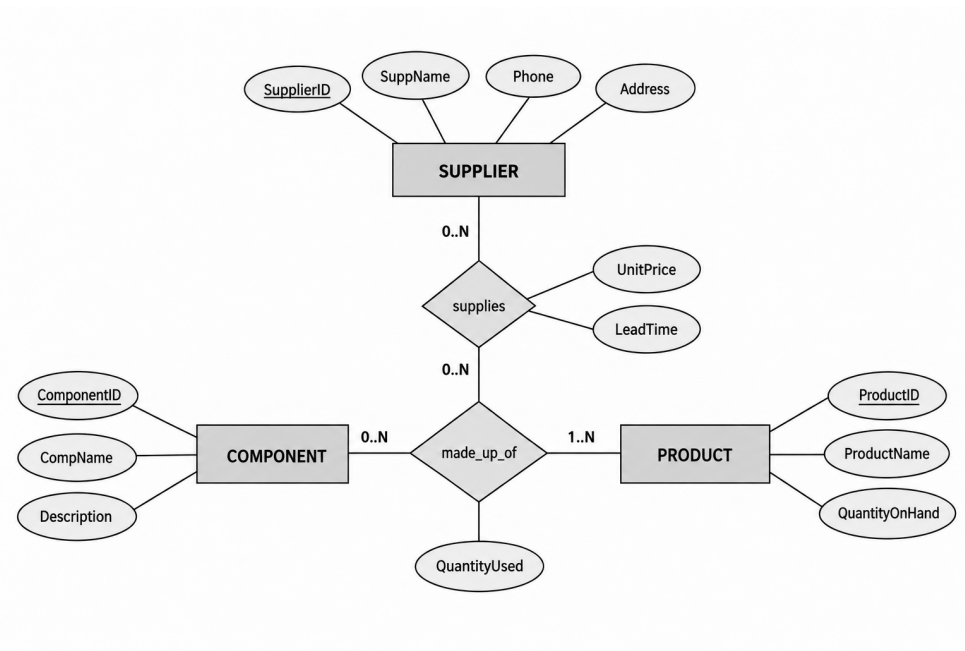


DBMS LAB ASSIGNMENT - 3

R.Thirumal
24MAB0A25

Q1 — Assumptions

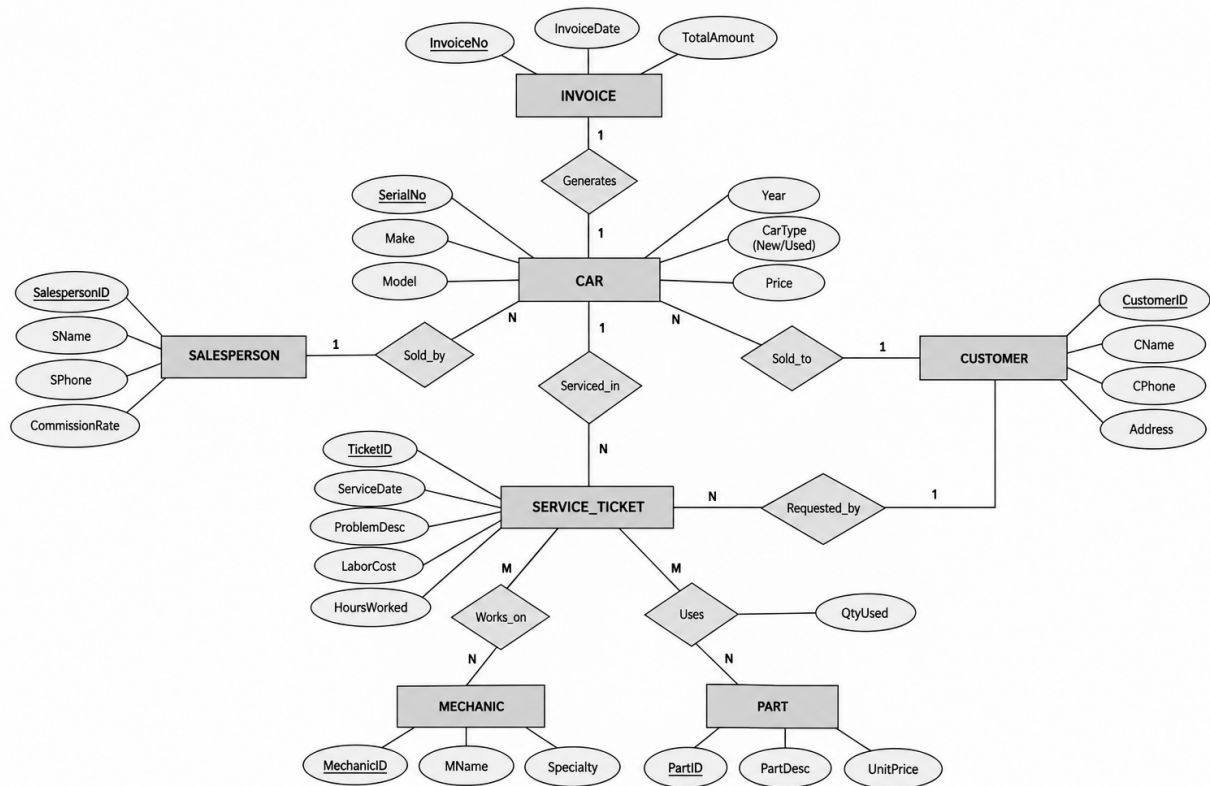
1. Each Product is uniquely identified by ProductID.
2. Each Component is uniquely identified by ComponentID.
3. Each Supplier is uniquely identified by SupplierID.
4. A product must contain at least one component, but a component may exist without being used in any product.
5. A component may be supplied by zero or more suppliers, and a supplier may supply zero or more components.
6. The combination of ProductID and ComponentID uniquely identifies a product-component relationship.
7. QuantityUsed represents the quantity of a particular component required for a particular product.
8. The combination of SupplierID and ComponentID uniquely identifies a supplier-component relationship.
9. UnitPrice and LeadTimeDays are attributes of the supplier–component relationship, since the price and lead time may differ between suppliers.
10. A supplier can exist without supplying any component.
11. Component and product IDs are assumed to be non-null and unique.



PRODUCT (<u>ProductID</u> , ProductName, QuantityOnHand) PK: ProductID	COMPONENT (<u>ComponentID</u> , ComponentName, Description) PK: ComponentID
SUPPLIER (<u>SupplierID</u> , SupplierName, ContactPhone, Address) PK: SupplierID	PRODUCT_COMPONENT (<u>ProductID</u> , <u>ComponentID</u> , QuantityUsed) PK: (ProductID, ComponentID) FK: ProductID → PRODUCT (ProductID) FK: ComponentID → COMPONENT (ComponentID)
SUPPLIER_COMPONENT (<u>SupplierID</u> , <u>ComponentID</u> , UnitPrice, LeadTimeDays) PK: (SupplierID, ComponentID) FK: SupplierID → SUPPLIER (SupplierID) FK: ComponentID → COMPONENT (ComponentID)	

Q2 — Assumptions

1. Each car is uniquely identified by SerialNo.
2. A car may initially exist as unsold inventory, so SalespersonID and CustomerID can be NULL until the car is sold.
3. Once a car is sold, it has exactly one salesperson.
4. Once a car is sold, it has exactly one customer.
5. A salesperson can sell zero, one, or many cars.
6. A customer can purchase zero, one, or many cars.
7. Each car sale generates exactly one invoice.
8. Each invoice belongs to exactly one car sale, so SerialNo in INVOICE is NOT NULL and UNIQUE.
9. A customer can use the dealership's service facility without purchasing a car.
10. Each service visit creates a separate Service Ticket, uniquely identified by TicketID.
11. Every service ticket is associated with exactly one car, identified by SerialNo.
12. A car can have zero, one, or many service tickets, allowing its complete service history to be maintained.
13. A service ticket may involve zero, one, or many mechanics, and a mechanic may work on zero, one, or many service tickets.
14. The combination of TicketID and MechanicID uniquely identifies a WORKS_ON record.
15. A service ticket may use zero, one, or many replacement parts, while a part may be used on many service tickets.
16. The combination of TicketID and PartID uniquely identifies a USES record.
17. HoursWorked is an attribute of the service-ticket–mechanic relationship.
18. QtyUsed is an attribute of the service-ticket–part relationship.



SALESPERSON (<u>SalespersonID</u> , SName, SPhone, CommissionRate) PK: SalespersonID	CUSTOMER (<u>CustomerID</u> , CName, CPhone, CAddress) PK: CustomerID
MECHANIC (<u>MechanicID</u> , MName, Specialty) PK: MechanicID	PART (<u>PartID</u> , PartDesc, UnitPrice) PK: PartID
CAR (<u>SerialNo</u> , Make, Model, Year, CarType, Price, SalespersonID, CustomerID) PK: SerialNo FK: SalespersonID → SALESPERSON (SalespersonID) [nullable — unsold car] FK: CustomerID → CUSTOMER (CustomerID) [nullable — unsold car]	INVOICE (<u>InvoiceNo</u> , InvoiceDate, TotalAmount, SerialNo) PK: InvoiceNo FK: SerialNo → CAR (SerialNo) [not null, unique — one invoice per car sale]
SERVICE_TICKET (<u>TicketID</u> , ServiceDate, ProblemDesc, LaborCost, SerialNo, CustomerID) PK: TicketID FK: SerialNo → CAR (SerialNo) FK: CustomerID → CUSTOMER (CustomerID)	WORKS_ON (<u>TicketID</u> , <u>MechanicID</u> , HoursWorked) PK: (TicketID, MechanicID) FK: TicketID → SERVICE_TICKET (TicketID) FK: MechanicID → MECHANIC (MechanicID)
USES (<u>TicketID</u> , <u>PartID</u> , QtyUsed) PK: (TicketID, PartID) FK: TicketID → SERVICE_TICKET (TicketID) FK: PartID → PART (PartID)	